SOP for Automation and Monitoring of Conveyor and Boiler Systems Using PLC & SCADA

Purpose

The purpose of this SOP is to provide standardized, step-by-step instructions for the safe and efficient automation, operation, and monitoring of the conveyor and boiler systems within the thermal power plant. This SOP ensures that all electrical and electronic components, including sensors, relays, motors, PLCs, and SCADA systems, are correctly installed, configured, and operated to maintain plant efficiency, ensure personnel safety, and prevent equipment damage. It also provides guidance for troubleshooting common operational issues and maintaining compliance with IEC standards (IEC 61131 for PLC and IEC 60870-5 for SCADA).

2. Scope

This SOP applies to the automation, operation, monitoring, and troubleshooting of conveyor and boiler systems using PLC and SCADA. It covers the installation, configuration, and operation of electrical and electronic components, including sensors, relays, motors, and control logic, to ensure efficient material handling, process regulation, and safe operation of the systems. This SOP is intended for engineers, technicians, and personnel responsible for system setup, maintenance, and process control.

3. Safety & Precautions

High-Risk Areas:
Electrical panels and PLC control cabinets (risk of high voltage).
Boiler and tank systems (risk of high temperature and pressurized steam).
Conveyor systems and moving mechanical parts (risk of entanglement or crushing).

Personal Protective Equipment (PPE):
Insulated gloves and safety shoes for electrical work.
Heat-resistant gloves and face shield when working near boilers or tanks.
Safety helmets and protective clothing when working near moving machinery.

Technical Safety Measures:

1. Ensure all power supplies to PLC, SCADA, and sensors are **isolated** before installation or maintenance.
2. Use **lockout-tagout (LOTO)** procedures on high-voltage equipment.
3. Verify **grounding and earthing** of all electrical panels and motor circuits.
4. Use relays and PLC interlocks to prevent overheating, overpressure, or conveyor overload.
5. Install **emergency stop switches** on conveyors and critical machinery.
6. Ensure ultrasonic and float-level sensors are mounted to avoid direct contact with hot fluids or corrosive materials.

4. Tools, Equipment & Materials Required

Equipment & Components:

PLC (Programmable Logic Controller): For automation and control of conveyors, boiler, pumps, and valves.

SCADA System & HMI: For real-time monitoring, data logging, and remote operation of the automated system.

Sensors:

1. Ultrasonic level sensors (for fuel/oil or water level detection).
2. Float switches (for liquid level detection in tanks).
3. Temperature sensors (RTDs or thermocouples for boiler monitoring).
4. Pressure sensors (for boiler and pipeline pressure monitoring).

Relays & Relay Drivers: For switching high-current loads based on PLC signals.

Motors: DC/AC motors for conveyor operation.

Actuators: Solenoid valves or motor-driven flaps for boiler and material handling regulation.

Cables & Connectors: Copper wires for signal and power transmission, optionally optical fiber for low-latency communication.

Software & Programming Tools:

1. PLC programming software (ladder logic or function block diagram).
2. SCADA configuration and HMI development tools.
3. Diagnostic and simulation software for sensor calibration and signal testing.

Installation & Troubleshooting Tools:

1. Multimeter and clamp meter (for voltage, current, and continuity checks).
2. Oscilloscope (for signal verification in sensors and relay circuits).

3. Screwdrivers, pliers, wire strippers, and crimping tools.
4. Insulation tester (megger) for safety checks on electrical circuits.

5. Procedure / Step-by-Step Instructions

Step 1: Component Installation

1. Mount all **sensors** (ultrasonic, float switch, temperature, and pressure sensors) securely in their designated locations on tanks, boilers, and conveyors. Ensure ultrasonic sensors have a clear path to the liquid surface for accurate level detection.
2. Install **motors** for the conveyors and connect them to motor starters or contactors.
3. Connect **relays** and relay drivers for high-current switching of heaters, pumps, and valves.
4. Route **wiring** from sensors, relays, and motors to the PLC input/output terminals, ensuring proper separation of **signal and power cables** to avoid interference.

Step 2: PLC & SCADA Setup

Power on the PLC and SCADA system. Ensure proper **grounding and isolation**.
Configure PLC I/O mapping:

- Inputs: Sensor signals (level, temperature, pressure, switch status)
- Outputs: Motor starters, heater relays, alarm signals

Configure SCADA HMI:

- Display real-time process values (temperature, pressure, liquid levels)
- Include control buttons for conveyors, heaters, and pumps
- Set alarm thresholds for over-temperature, overpressure, and low/high-level conditions

Step 3: Ladder Logic Programming

Write ladder logic for automated control:
Conveyor control: Start/stop motors based on feed/discharge requirements and interlocked sensors.
Boiler automation: Turn off heater if temperature exceeds threshold; regulate feed pump based on water level.
Pressure control: Open valve if pressure exceeds limit.
Implement **safety interlocks**: Ensure motors, heaters, and pumps are deactivated in case of sensor failure or emergency stop.

Step 4: System Testing & Calibration

1. Test each sensor for accurate readings. Calibrate ultrasonic and float-level sensors as per tank dimensions.
2. Verify PLC outputs by simulating inputs and checking actuator operation (motor starts/stops, relay activation).
3. Confirm SCADA HMI displays correct real-time data and triggers alarms correctly.
4. Test conveyor sequences:
5. Start one feeding conveyor → ensure corresponding discharge conveyor operates.
6. Check interlocking logic with lamps or indicators.

Step 5: Operation & Monitoring

1. Monitor the system through SCADA: temperature, pressure, and level readings should remain within set limits.
2. Observe conveyor operation: Ensure continuous material flow without jams.
3. Verify relay and motor operations: Heaters, pumps, and valves respond to PLC commands automatically.
4. Document readings, events, and alarms in logbooks or SCADA logs.

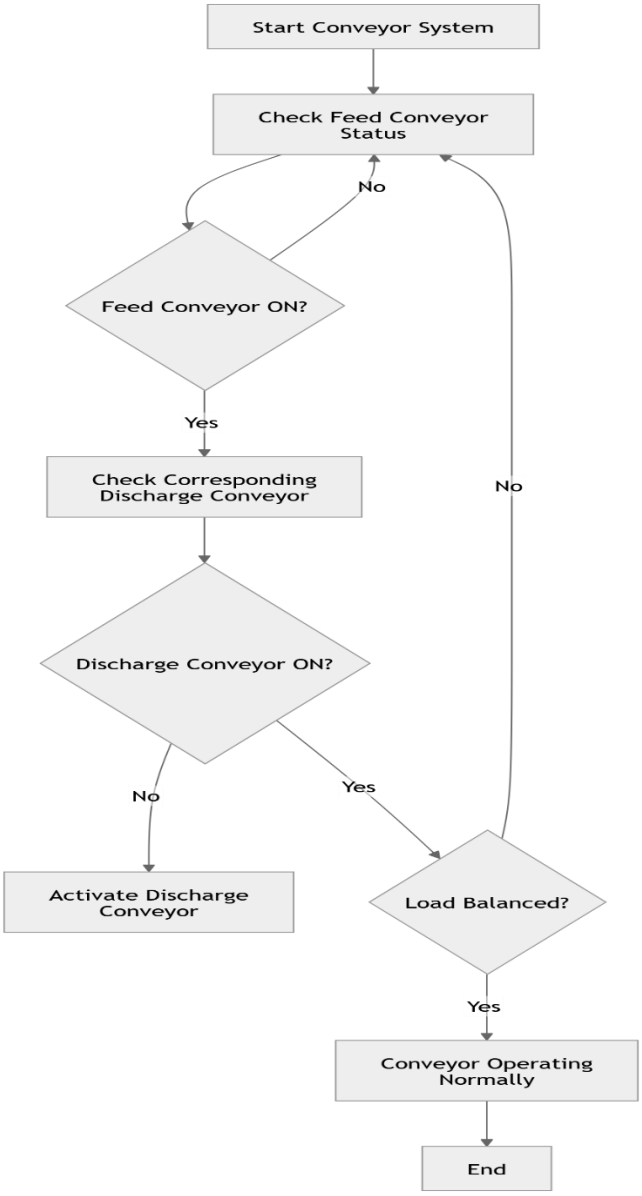
Step 6: Troubleshooting

If heaters or motors fail to operate, check:

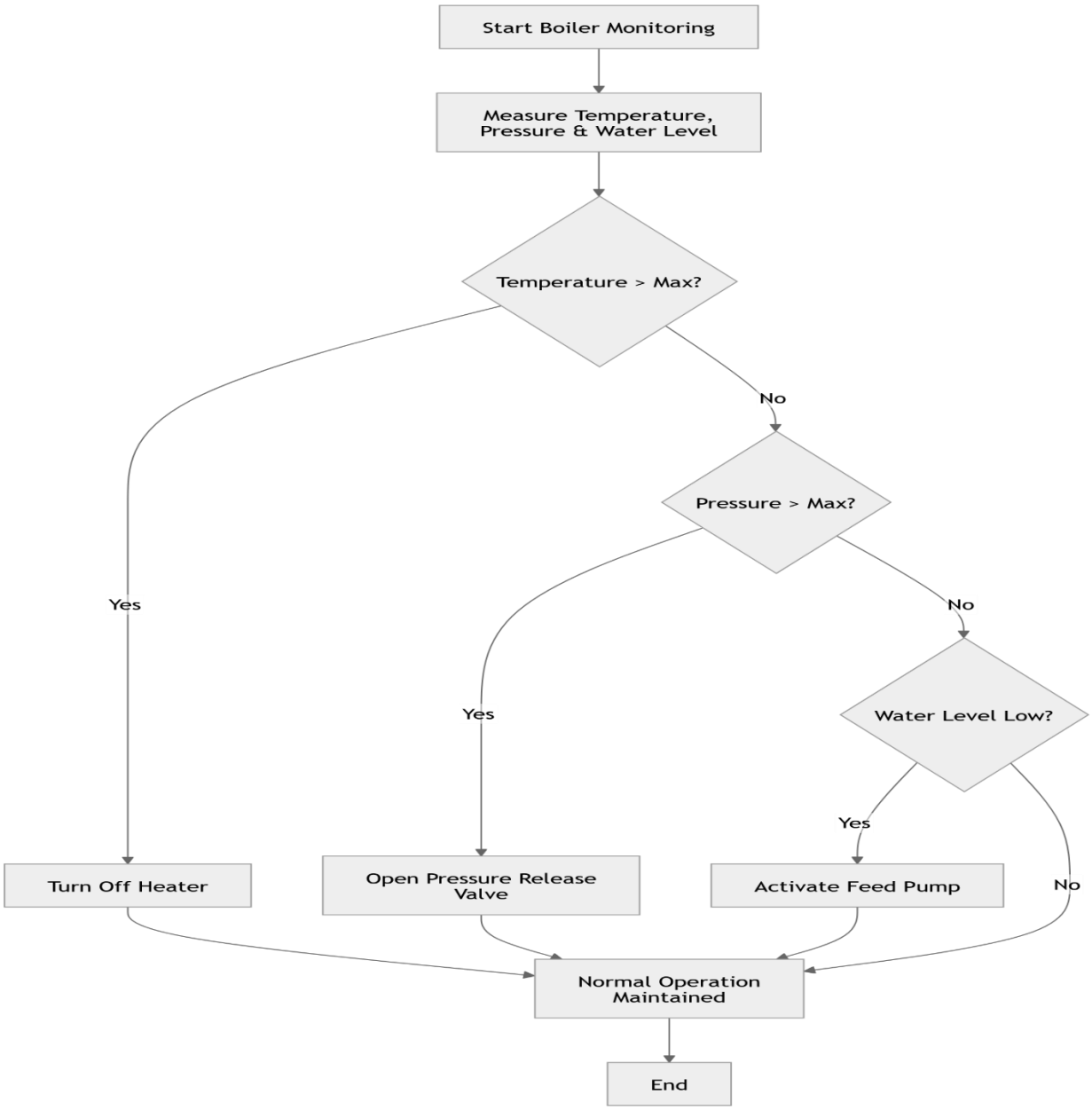
- Sensor input signals to PLC
- Relay and relay driver connections
- Ladder logic and SCADA mapping

If ultrasonic level readings are unstable, verify sensor alignment, frequency, and absence of obstructions.

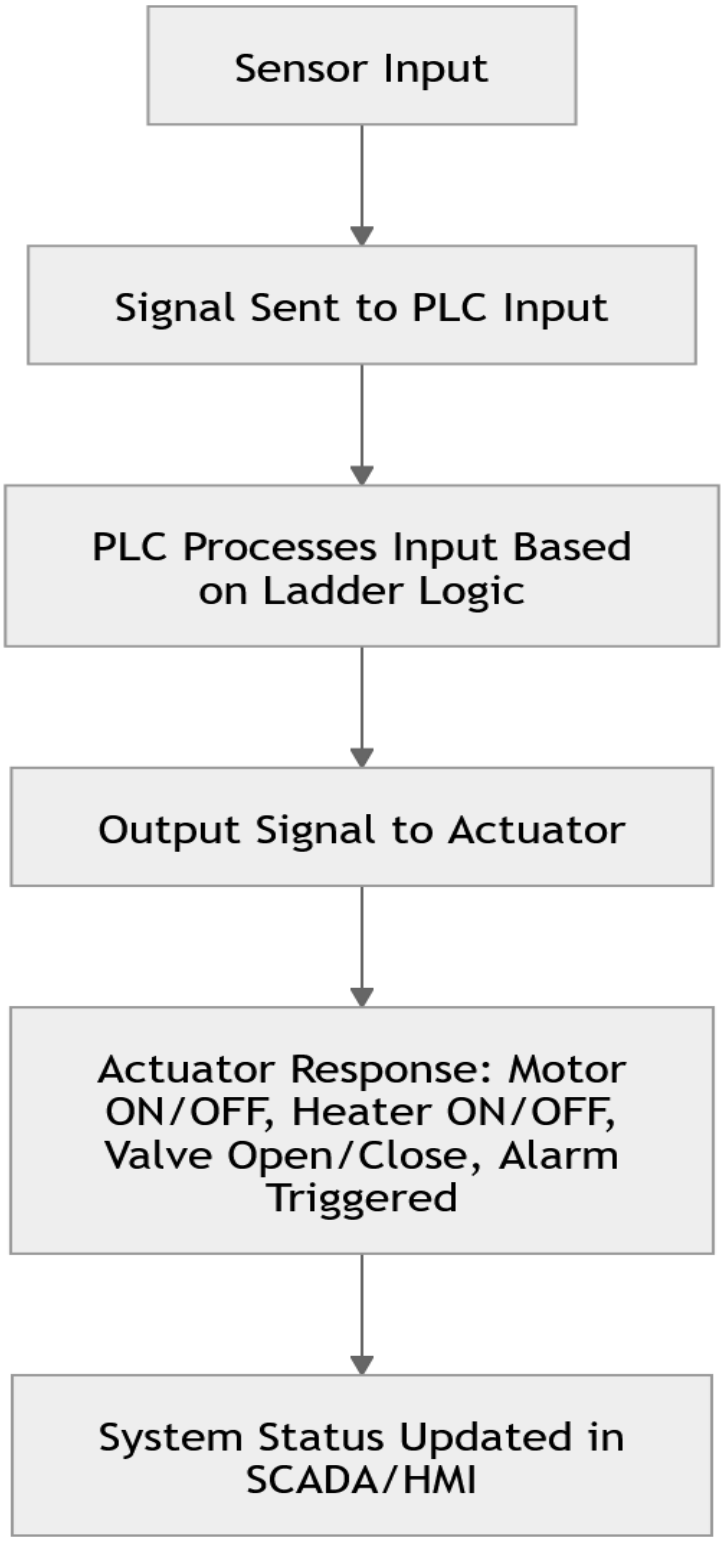
For conveyor jams, check motor connections, switch interlocks, and load balancing between feeding and discharge conveyors.



Conveyor Feed/Discharge Logic Flowchart



Boiler Temperature & Pressure Regulation Flowchart



Sensor → PLC → Actuator Signal Flowchart

6. Troubleshooting / Problem-Solving

This section provides **technical solutions for common operational issues** encountered in automated conveyor and boiler systems. Each problem includes the **cause, technical diagnosis, and corrective action**.

1. Heater Fails to Turn Off at High Temperature

Symptom: Heater remains ON even when temperature exceeds set threshold.
Technical Cause: PLC output signal magnitude too low to activate the relay.
Solution:

- Add a **relay driver circuit** between PLC output and heater relay.
- Ensure transistor amplifier boosts signal to required relay activation voltage/current.
- Verify relay operation using multimeter or oscilloscope to confirm switching.

Monitoring: Temperature readings on SCADA; confirm heater deactivates when threshold is reached.

2. Pressure Relief Valve Not Operating

Symptom: Boiler pressure exceeds set limit; valve fails to open.
Technical Cause: Incorrect PLC output mapping or faulty relay.
Solution:

- Check PLC ladder logic for correct I/O assignment for valve control.
- Inspect wiring and relay operation; replace faulty components if needed.
- Test valve manually to ensure actuator moves freely.

Monitoring: Pressure sensor readings; alarm on SCADA if pressure exceeds safe limits.

3. Ultrasonic or Float Sensor Gives Erratic Level Readings

Symptom: Tank level readings fluctuate or remain constant despite actual changes.
Technical Cause: Misalignment, obstruction, or wrong frequency range of ultrasonic sensor; float switch mechanical blockage.
Solution:

- Align ultrasonic sensor perpendicular to liquid surface, free from obstacles.
- Set operating frequency between 20–200 kHz; adjust gain if needed.
- Clean float switch and ensure free movement.
- Test signals at PLC input for proper voltage levels.

Monitoring: Verify readings on SCADA against manual measurement.

4. Conveyor Motor Does Not Start or Stops Unexpectedly

Symptom: Conveyor fails to move or intermittently stops.
Technical Cause:

- Motor overload or thermal trip.
- Faulty limit switch or interlock in PLC logic.
- Loose or damaged wiring.

Solution:

- Inspect motor current and voltage using clamp meter.
- Check limit switches, interlocks, and PLC ladder logic for proper sequence.
- Tighten connections and inspect for damaged cables.
- Ensure load is balanced between feeding and discharging conveyors.

Monitoring: Motor status on SCADA; check indicator lamps.

5. Ash Accumulation in Hopper / Material Not Moving

Symptom: Ash or material remains in hopper, blocking conveyor operation.
Technical Cause: Manual removal difficult; hydraulic or pneumatic displacement not installed.
Solution:

- Install a **high-pressure water pump** connected to plunger system.
- Use crankshaft-driven piston to generate sufficient pressure to displace ash.
- Verify continuous flow and empty hopper readiness before next batch.

Monitoring: Level sensors on hopper; confirm movement through SCADA indicators.

6. Relay or Actuator Not Responding to PLC Commands

Symptom: Relays, valves, or motors do not operate according to PLC logic.
Technical Cause: Signal attenuation, component failure, or incorrect wiring.
Solution:

- Measure PLC output voltage/current; ensure it meets actuator/relay requirements.
- Use relay driver or amplifier circuit to strengthen signal.
- Inspect wiring, connectors, and mechanical actuation.
- Confirm proper ladder logic sequence and I/O mapping in PLC.

Monitoring: Confirm response of actuators on SCADA; verify event logs.

7. SCADA/HMI Display Issues

Symptom: Incorrect readings, frozen interface, or missing alarms.
Technical Cause: Communication failure between PLC and SCADA, cable issues, or software misconfiguration.
Solution:

- Check communication cables and terminate properly.
- Verify SCADA configuration, including polling intervals and tag mapping.
- Reboot SCADA HMI system if necessary.

Monitoring: Verify live updates, alarms, and event logs in SCADA.

7. References / Standards

The following standards and technical references are used to ensure compliance, interoperability, and safe operation of the automated system:

Industry Standards

- **IEC 61131 (Programmable Logic Controllers):**
  Defines standards for PLC programming languages (Ladder Diagram, Function Block Diagram), hardware architecture, I/O configuration, and communication. Ensures reliable and standardized control logic implementation.
- **IEC 60870-5 (SCADA Communication Protocol):**
  Specifies communication protocols for telecontrol systems used in SCADA. Ensures reliable data transmission between PLCs, RTUs, and SCADA systems for real-time monitoring and control.
- **IEC 60204-1 (Electrical Safety of Machinery):**
  Covers safety requirements for electrical equipment in industrial machines, including wiring, grounding, and protection against electrical hazards.
- **IEC 60529 (Ingress Protection - IP Ratings):**
  Defines protection levels of enclosures against dust and water for sensors, control panels, and field devices.

Technical Documentation

PLC Manuals & Datasheets:
Provide details on I/O specifications, voltage/current ratings, memory capacity, and programming instructions.
SCADA/HMI Configuration Guides:
Include setup procedures for communication protocols, tag mapping, alarm configuration, and data logging.
Sensor Datasheets:
Ultrasonic sensors: Operating frequency (20–200 kHz), range, accuracy.
Temperature sensors (RTD/Thermocouple): Measurement range and calibration.
Pressure sensors: Pressure limits and output signal type (analog/digital).
Motor & Relay Specifications:
Define rated voltage, current capacity, switching characteristics, and protection requirements.

Best Practices & Internal Procedures

- Standard wiring practices for **signal and power separation** to minimize noise.
- Proper **grounding and shielding** for accurate sensor readings.
- Periodic **calibration and testing** of sensors and control systems.
- Documentation of **PLC logic changes and SCADA configurations** for traceability.

8. Record-Keeping / Logs

Proper record-keeping ensures traceability, system reliability, and effective troubleshooting of automated operations.

Operational Data Logging

- Record real-time process parameters through SCADA, including:
- Temperature, pressure, and liquid level readings
- Conveyor status (ON/OFF, load conditions)
- Motor current and voltage levels
- Configure SCADA for **historical data logging (trending)** with defined sampling intervals.
- Store logs in a centralized database or local server for analysis and audit purposes.

Event & Alarm Logs

- Maintain automatic logs of:
- Alarm conditions (over-temperature, overpressure, low/high level)
- Equipment faults (motor trips, relay failures, sensor errors)
- Emergency stop activations
- Ensure time-stamped entries for accurate fault diagnosis and sequence-of-events analysis.

Maintenance & Calibration Records

- Document all maintenance activities, including:
- Sensor calibration (ultrasonic, temperature, pressure sensors)
- Relay, motor, and actuator inspections
- Replacement of faulty components
- Include test results, calibration values, and technician notes.

PLC & SCADA Configuration Logs

- Maintain version-controlled records of:
- PLC ladder logic updates and modifications
- SCADA tag configuration and alarm setpoint changes
- Record date, change description, and responsible personnel for traceability.

Troubleshooting & Incident Reports

- Log all system issues with:
- Observed symptoms and affected components
- Root cause analysis
- Corrective actions taken (e.g., relay driver installation, wiring correction)

- Use logs to identify recurring faults and improve system reliability.

Importance of Record-Keeping

- Enables **predictive maintenance** and reduces downtime.
- Supports **compliance with industrial standards and audits**.
- Improves **system performance analysis and optimization**.
- Ensures **accountability and knowledge transfer** for future operations.

9. Definitions / Abbreviations

PLC (Programmable Logic Controller): Industrial digital controller used for automation of electromechanical processes.
SCADA (Supervisory Control and Data Acquisition): System used for real-time monitoring, control, and data acquisition of industrial processes.
HMI (Human-Machine Interface): Interface that allows operators to interact with the control system (visualization, control inputs).
RTD (Resistance Temperature Detector): Sensor used to measure temperature based on resistance variation.
Relay: Electrically operated switch used to control high-power circuits using low-power signals.
Actuator: Device (motor, valve, etc.) that performs physical action based on control signals.
Ultrasonic Sensor: Sensor that measures distance/level using high-frequency sound waves.
Ladder Logic: Graphical programming language used for PLC programming.

10. Roles & Responsibilities

Control Engineer:

- Design and configure PLC logic and SCADA system.
- Ensure proper integration of sensors and actuators.

Maintenance Technician:

- Perform routine inspection, calibration, and repair of sensors, relays, and motors.
- Troubleshoot electrical and automation faults.

Operator:

- Monitor system through HMI/SCADA.
- Respond to alarms and ensure normal operation of conveyors and boiler systems.

Supervisor/Engineer-in-Charge:

- Approve system changes and ensure compliance with safety and operational standards.
- Review logs and performance reports.